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Assignment - I Recall Analysis

Question 1

$$\text{pred}(0) := 0$$

$$\text{pred}(n++) := n$$

$$m \ominus 0 := m$$

$$m \ominus n++ := \text{pred}(m \ominus n)$$

(a) We can use induction.

$$P(n): \quad \text{~~0~~ } 0 \ominus n = 0$$

Base case:

$$P(0): \quad 0 \ominus 0 = 0$$

$$0 = 0$$

[definition]

$\therefore P(0)$ holds

Induction hypothesis: Let, $P(n)$ is true.

Inductive step:

$$P(n++): \quad 0 \ominus n++ = 0$$

$$\text{pred}(0 \ominus n) = 0$$

$$\text{pred}(0) = 0$$

$$0 = 0$$

[definition]

[induction hypothesis]

$$\therefore P(n) \rightarrow P(n++)$$

$$\therefore 0 \ominus n = 0 \quad \forall n \in \mathbb{N}$$

(b) We can use induction.

$$P(n): m++ \ominus n++ = m \ominus n \quad m \in \mathbb{N}$$

Base case:

$$P(0): m++ \ominus 0++ = m \ominus 0$$

$$\text{pred}(m++ \ominus 0) = m$$

$$\text{pred}(m++) = m$$

$$m = m$$

[definition]

[definition]

[definition]

$\therefore P(0)$ holds

Induction hypothesis: Let, $P(n)$ is true.

Inductive step:

$$P(n++): m++ \ominus (n++)++ = m \ominus n++$$

$$\text{pred}(m++ \ominus n++) = \text{pred}(m \ominus n) \quad [\text{def}^n]$$

$$\text{pred}(m \ominus n) = \text{pred}(m \ominus n) \quad [\text{induction hypothesis}]$$

$$\therefore P(n) \rightarrow P(n++)$$

$$\therefore m++ \ominus n++ = m \ominus n \quad \forall n, m \in \mathbb{N}$$

(c) We can use induction.

$$P(n): (m \ominus n) \ominus k = m \ominus (n+k) \quad m, k \in \mathbb{N}$$

Base case:

$$P(0): (m \ominus 0) \ominus k = m \ominus (0+k)$$

$$\text{LHS} = (m \ominus 0) \ominus k$$

$$= m \ominus k$$

[definition]

(c) We can use induction.

$$P(k): (m \ominus n) \ominus k = m \ominus (n+k) \quad m, n \in \mathbb{N}$$

Base case:

$$P(0): (m \ominus n) \ominus 0 = m \ominus (n+0) \\ m \ominus n = m \ominus n$$

$\therefore P(0)$ holds

Induction hypothesis: Let, $P(k)$ is true

Inductive step:

$$P(k++) \Rightarrow (m \ominus n) \ominus k++ = m \ominus (n+k++)$$

$$\begin{aligned} \text{LHS} &= (m \ominus n) \ominus k++ \\ &= \text{pred}((m \ominus n) \ominus k) && [\text{definition}] \\ &= \text{pred}(m \ominus (n+k)) && [\text{induction hypothesis}] \end{aligned}$$

$$\begin{aligned} \text{RHS} &= m \ominus (n+k++) \\ &= m \ominus (n+k)++ \\ &= \text{pred}(m \ominus (n+k)) && [\text{definition}] \end{aligned}$$

$$\therefore P(k) \rightarrow P(k++)$$

$$\therefore (m \ominus n) \ominus k = m \ominus (n+k) \quad \forall m, n, k \in \mathbb{N}.$$

(d) We can use induction

$$P(n): (m+n) \ominus n = m \quad m \in \mathbb{N}$$

Base case:

$$P(0): m \oplus (m+0) \ominus m = m$$

$$m \oplus 0 = m$$

$$m = m$$

[definition]

$\therefore P(0)$ holds

Induction hypothesis: Let, $P(n)$ is true.

Inductive step:

$$P(n++): (m+n++) \ominus n++ = m$$

$$\text{LHS} = (m+n)++ \ominus n++$$

$$= (m+n) \ominus n$$

[previously proved]

$\therefore P(n++)$ is $(m+n) \ominus n = m$ which is induction hypothesis.

$$\therefore P(n) \rightarrow P(n++)$$

$$\therefore (m+n) \ominus n = m \quad \forall m, n \in \mathbb{N}$$

(e) Proving lemma $m \oplus m = 0 \quad \forall m \in \mathbb{N}$:

$$P(m): m \oplus m = 0$$

$$P(0): 0 \oplus 0 = 0 \quad \text{holds}$$

$$P(m++): m++ \oplus m++ = 0$$

$$m \oplus m = 0$$

which is $P(m)$

$$\therefore P(0) \text{ and } P(m) \rightarrow P(m++)$$

$\therefore P(m)$ holds $\forall m \in \mathbb{N}$ by principle of mathematical induction.

Proving $m \leq n \Rightarrow m \ominus n = 0,$

$$m \leq n \Rightarrow n = m + k \quad k \in \mathbb{N}$$

$$\begin{aligned} \therefore m \ominus n &= m \ominus (m + k) \\ &= (m \ominus m) \ominus k \\ &= 0 \ominus k \\ &= 0 \end{aligned}$$

[previously proved]

[previously proved]

[previously proved]

$$\therefore m \leq n \Rightarrow m \ominus n = 0$$

Proving $m \ominus n = 0 \Rightarrow m \leq n,$

Let, $m > n.$

Then, $m = n + k \quad k \in \mathbb{N}, k \neq 0$

$$\therefore m \ominus n = (n + k) \ominus n = k \neq 0$$

But that ~~is not~~ is a contradiction.

$$\therefore m \neq n$$

$$\therefore m \leq n$$

$$\therefore m \ominus n = 0 \Rightarrow m \leq n$$

Implication is proved in both directions

$$\therefore m \leq n \Leftrightarrow m \ominus n = 0$$

(f) Proving $n \leq m \Rightarrow (m \ominus n) + n = m$,

$$n \leq m \Rightarrow m = n + k \quad k \in \mathbb{N}$$

$$\begin{aligned} \therefore (m \ominus n) + n &= ((n+k) \ominus n) + n \\ &= k + n \\ &= m \end{aligned}$$

[previously proved]

$\therefore$ proved

Proving $(m \ominus n) + n = m \Rightarrow n \leq m$,

By trichotomy, either $n < m$ or $n \geq m$,

If $n \geq m$, $n = m + k$, $k \in \mathbb{N}$.

$$\begin{aligned} (m \ominus n) + n &= 0 + n \\ &= m + k \end{aligned}$$

[previously proved]

$$\text{But } (m \ominus n) + n = m.$$

$$\therefore k = 0$$

$$\therefore n = m$$

$$\therefore n < m \text{ or } n = m$$

$$\therefore n \leq m$$

$$\therefore (m \ominus n) + n = m \Rightarrow n \leq m$$

$$\therefore (m \ominus n) + n = m \Leftrightarrow n \leq m$$

$$(g) \quad \max(m, n) := \begin{cases} m, & n \leq m \\ n, & \text{otherwise} \end{cases}$$

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Case 1: $n \leq m$

$$n \ominus m = 0$$

$$m + (n \ominus m) = m$$

[previously proved]

And, we know $\max(m, n) = m$

Case 2: $m > n$

Then, $n = m + k$ $k \in \mathbb{N}, k \neq 0$

$$\text{Then, } n \ominus m = (m + k) \ominus m = k$$

$$\text{Then, } m + (n \ominus m) = m + k = n$$

And, $\max(m, n) = n$

$\therefore \max(m, n) = m + (n \ominus m)$ [in all cases]

Since $\max$ is symmetric, we can replace n and m ,

$$\begin{aligned} \max(n, m) &= n + (m \ominus n) = \max(m, n) \\ &= m + (n \ominus m) \end{aligned}$$

$\therefore$ proved

(h) We can use induction.

$$P(n): k \times (m \ominus n) = (k \times m) \ominus (k \times n)$$

Base case:

$$P(0): k \times (m \ominus 0) = (k \times m) \ominus (k \times 0)$$

(h) By induction, either $m \leq n$ or $m > n$.

Case 1: $m \leq n$

$$\begin{aligned} \text{LHS} &= k \times (m \ominus n) \\ &= k \times 0 \\ &= 0 \end{aligned} \quad [\text{previously proved}]$$

$$\text{RHS} = (k \times m) \ominus (k \times n)$$

$$m \leq n \Rightarrow k \times m \leq k \times n$$

$$\therefore \text{RHS} = 0$$

$$\therefore \text{LHS} = \text{RHS}$$

Case 2: $m > n$

$$\text{Then, } m = n + h \quad h \in \mathbb{N}, h \neq 0$$

$$\begin{aligned} \text{LHS} &= k \times ((n+h) \ominus n) \\ &= k \times h \end{aligned} \quad [\text{previously proved}]$$

$$\begin{aligned} \text{RHS} &= (k \times (n+h)) \ominus (k \times n) \\ &= (k \times n + k \times h) \ominus (k \times n) \\ &= k \times h \end{aligned} \quad [\text{previously proved}]$$

$$\therefore \text{LHS} = \text{RHS}$$

$$\therefore k \times (m \ominus n) = (k \times m) \ominus (k \times n)$$

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(i) To prove: $(k \leq m) \wedge (k \leq n) \rightarrow (m \ominus k = n \ominus k \Rightarrow m = n)$

Proof:

$$k \leq m \Rightarrow m = k + d \quad d \in \mathbb{N}$$

$$k \leq n \Rightarrow n = k + e \quad e \in \mathbb{N}$$

Now, $m \ominus k = n \ominus k$

$$\star (k+d) \ominus k = (k+e) \ominus k$$

$$d = e$$

$$k+d = k+e$$

$$m = n$$

$\therefore m \ominus k = n \ominus k \Rightarrow m = n$ if $k \leq m$ and $k \leq n$.

Both hypothesis are required.

Take $n=1, k=3, m=3$.

Now, $k \leq m$ but $k \not\leq n$.

$$\star m \ominus k = 0 = n \ominus k$$

But $n \neq m$.

Similar example can be shown for $k \leq m$ and $k \leq n$.

(2)

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Problem 3:

We can use induction.

$$P(k): (a_1 + \dots + a_n)^n \geq n^n a_1 \dots a_n \quad n = 2^k$$

Base case:

$$P(0): \begin{array}{l} (a_1)^{2^0} \geq (2^0)^{2^0} a_1 \\ a_1 \geq a_1 \end{array}$$

$\therefore P(0)$ holds

Induction hypothesis: Let, $P(k)$ is true.

Inductive step:

We need to prove $P(k+1)$

$$\text{Number of variables} = 2^{k+1} = 2 \times 2^k$$

We can group them as $a_1, \dots, a_n$ and $a_{n+1}, \dots, a_{2n}$ to use our induction hypothesis.

$$A := a_1 + \dots + a_n$$

$$B := a_{n+1} + \dots + a_{2n}$$

By induction hypothesis,

$$A^n \geq n^n a_1 \dots a_n \quad \text{--- ①}$$

$$B^n \geq n^n a_{n+1} \dots a_{2n} \quad \text{--- ②}$$

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We know about elementary identity:

$$(A+B)^2 \geq 4AB$$

Then, $(A+B)^{2^n} \geq 4^n A^n B^n$

$$\therefore (A+B)^{2^n} \geq 4^n (n^n a_1 \dots a_n) (n^n a_{n+1} \dots a_{2n})$$

$$(A+B)^{2^n} \geq 2^{2^{k+1}} (2^k)^{2^{k+1}} a_1 \dots a_{2^{k+1}}$$

$$(A+B)^{2^{k+1}} \geq (2^{k+1})^{2^{k+1}} a_1 \dots a_{2^{k+1}}$$

Which is $P(k+1)$. $\therefore P(k) \rightarrow P(k+1)$

$$\therefore (a_1 + \dots + a_{2^k})^{2^k} \geq (2^k)^{2^k} a_1 \dots a_{2^k} \quad \forall k \in \mathbb{N}.$$

~~(4)~~

Problem 4:

We know, $a_1 \leq a_2 \leq \dots \leq a_n$
 $b_1 \leq b_2 \leq \dots \leq b_n$

$$\therefore \text{if } i \leq j, \quad a_i \leq a_j \Rightarrow (a_i - a_j) \leq 0$$
$$b_i \leq b_j \Rightarrow (b_i - b_j) \leq 0$$

$$\therefore (a_i - a_j)(b_i - b_j) \geq 0$$

$$a_i b_i - a_i b_j - a_j b_i + a_j b_j \geq 0$$

$$a_i b_i + a_j b_j \geq a_i b_j + a_j b_i$$

Since this holds for every i and j , we can sum it over all i and j .

$$\sum_{i=1}^n \sum_{j=1}^n (a_i b_i + a_j b_j) \geq \sum_{i=1}^n \sum_{j=1}^n (a_i b_j + a_j b_i)$$

$$\text{LHS} = \sum_{i=1}^n \sum_{j=1}^n a_i b_i + \sum_{i=1}^n \sum_{j=1}^n a_j b_j$$

$$= n \sum_{i=1}^n a_i b_i + n \sum_{j=1}^n a_j b_j$$

$$= 2n \sum_{i=1}^n a_i b_i$$

$$\text{RHS} = \sum_{i=1}^n \sum_{j=1}^n a_i b_j + \sum_{i=1}^n \sum_{j=1}^n a_j b_i$$

$$= 2 \left(\sum_{i=1}^n a_i \right) \left(\sum_{j=1}^n b_j \right)$$

[a standard identity]

$$\therefore \sum_{i=1}^n a_i b_i \geq \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right) \quad [\text{after cancelling } 2]$$

Problem 5:

Assume $f: \mathbb{N} \rightarrow \mathbb{N}$ satisfying $f(f(n)) = n++ \quad \forall n \in \mathbb{N}$ exists.

Then, $f(f(f(n))) = f(n++)$

$$f(n++) = f(n)++$$

After some ~~rough~~ rough observations, we can find a pattern that,

$$\begin{aligned} f(1) &= f(0) + 1 \\ f(2) &= f(0) + 2 \end{aligned}$$

Let's formalize it using induction:

$P(n): f(n) = f(0) + n$ is our induction hypothesis

$P(0): f(0) = f(0)$ holds

$P(n++):$

$$\begin{aligned} f(n++) &= f(0) + n++ \\ f(n++) &= (f(0) + n)++ \\ f(n) &= f(0) + n \end{aligned}$$

which is our induction hypothesis.

$$\therefore P(n) \Rightarrow P(n++)$$

$$\therefore f(n) = f(0) + n \quad \text{is true } \forall n \in \mathbb{N}.$$

Then, $f(f(n)) = f(f(0) + n) = n++$

$$(f(0) + n) + f(0) = n + 1$$

$$f(0) + f(0) = 1$$

But, there is no $m \in \mathbb{N}$ st $m+m=1$.
 $\therefore$ no such function can exist.

Part 2:

We can define $f: \mathbb{N} \rightarrow \mathbb{N}$ st $f(n) := n++$

$$\text{Then, } f(f(n)) = f(n++) = \overline{f}(n++)++$$

$\therefore$ such a function can exist.

Problem 6:

Meaning of ' ' (informally) in the notation is indicating the presence of ^{iterated} powerset operation results.

To formally define such a set we can:

$$P^0(x) := x$$

$$P^n(x) := P(P^{n-1}(x)) \quad \text{where } P(A) \text{ is powerset of } A.$$

Then using axiom of replacement, the set

$$\{ \exists y : y = P^n(x) \text{ for some } n \in \mathbb{N} \} \text{ exists.}$$

Or more compactly, $\{ P^n(x) : n \in \mathbb{N} \}$ exists and is equal to $\{ x, P(x), P(P(x)), \dots \}$.

It should be noted, we haven't used recursion theorem but the concept of recursion alongside replacement axiom (which requires a property, not a function).

Problem 7:

Suppose, $f: X \rightarrow P(X)$ is surjective.

Define, $A := \{x \in X, x \notin f(x)\}$

Clearly, $A \subseteq X$ and hence, $A \in P(X)$.

Since f is surjective, every element of $P(X)$ must be given by f . That is, there must exist some $a \in X$ st $f(a) = A$.

Now, $a \in A \Leftrightarrow a \notin f(a) \Leftrightarrow a \notin A$.

This a contradiction.

$\therefore$ no surjective $f: X \rightarrow P(X)$ exists.

The method of contradiction (construction of a specific set) is similar among this problem and Russell's paradox.

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Problem 8:

Given: If $X \subseteq Y$, $\text{inc}_{X \rightarrow Y}: X \rightarrow Y$ is defined by

$$\text{inc}_{X \rightarrow Y}(x) := x \quad \forall x \in X$$

(a) Both functions have domain X and Range Z .

$$\begin{aligned} \text{For any } x \in X, \quad & \text{inc}_{Y \rightarrow Z} \circ \text{inc}_{X \rightarrow Y}(x) \\ &= \text{inc}_{Y \rightarrow Z}(\text{inc}_{X \rightarrow Y}(x)) \\ &= \text{inc}_{Y \rightarrow Z}(x) \\ &= x \end{aligned}$$

~~But~~ And $\forall x \in X$, $\text{inc}_{X \rightarrow Z}(x) = x$

$$\therefore \text{inc}_{Y \rightarrow Z} \circ \text{inc}_{X \rightarrow Y} = \text{inc}_{X \rightarrow Z}$$

$$\begin{aligned} (b) \quad f \circ \text{inc}_{A \rightarrow A}(x) &= f(\text{inc}_{A \rightarrow A}(x)) \quad [\text{inc}_{A \rightarrow A}(a) = a \quad \forall a \in A] \\ &= f(x) \quad \forall x \in A \end{aligned}$$

$$\begin{aligned} \text{inc}_{B \rightarrow B} \circ f(x) &= \text{inc}_{B \rightarrow B}(f(x)) \quad [\text{inc}_{B \rightarrow B}(b) = b \quad \forall b \in B] \\ &= f(x) \quad \forall x \in A \end{aligned}$$

Domain and range of ~~both~~ functions match.

$$\therefore f \circ \text{inc}_{A \rightarrow A} = \text{inc}_{B \rightarrow B} \circ f = f$$

(c) $f: A \rightarrow B$ is bijective

$$\therefore \text{if } f(a) = b, \quad f^{-1}(b) = a \quad \begin{matrix} a \in A \\ b \in B \end{matrix}$$

$$\therefore f \circ f^{-1}(x) = f(f^{-1}(x)) = x \quad \forall x \in B$$

And, $\text{inc}_{B \rightarrow B}(x) = x \quad \forall x \in B$

$$\text{Similarly, } f^{-1} \circ f(x) = f^{-1}(f(x)) = x \quad \forall x \in A$$

And, $\text{inc}_{A \rightarrow A}(x) = x \quad \forall x \in A$

$$\therefore f \circ f^{-1} = \text{inc}_{B \rightarrow B} \quad \text{and} \quad f^{-1} \circ f = \text{inc}_{A \rightarrow A}$$

(d) We can define, $h(u) := \begin{cases} f(u), & u \in X \\ g(u), & u \in Y \end{cases}$

Reasons - 1. This satisfies following conditions.
2. Cases on u are exhaustive and disjoint.

Now, for $x \in X$,

$$\begin{aligned} h \circ \text{inc}_{X \rightarrow X \cup Y}(x) &= h(\text{inc}_{X \rightarrow X \cup Y}(x)) \\ &= h(x) \\ &= f(x) \end{aligned}$$

$$\therefore h \circ \text{inc}_{X \rightarrow X \cup Y} = f$$

And, for $y \in Y$,

$$\begin{aligned} h \circ \text{inc}_{Y \rightarrow X \cup Y}(y) &= h(\text{inc}_{Y \rightarrow X \cup Y}(y)) \\ &= h(y) \\ &= g(y) \end{aligned}$$

$$\therefore h \circ \text{inc}_{Y \rightarrow X \cup Y} = g$$

Uniqueness proof:

Suppose another function $h': X \rightarrow Y$ satisfies same conditions.

Case 1: $u \in X$

~~$$h'(u) = h(u)$$~~

$$h'(\text{inc}_{X \rightarrow X \cup Y}(u)) = \text{inc}_{X \rightarrow X \cup Y}(u) = f(u)$$

$$h'(u) = f(u)$$

$$\therefore h' = h$$

Case 2: $u \in Y$

Similarly, $h'(\text{inc}_{Y \rightarrow X \cup Y}(u)) = g(u)$

$$h'(u) = g(u)$$

$$\therefore h' = h$$

$\therefore h$ is unique.

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Problem 2:

(a) We can use induction.

$$P(n): \quad n=0 \text{ or } n=0++ \text{ or } n=(m++)++ \text{ for some } m \in \mathbb{N}$$

Base case:

$$P(0): \quad 0=0$$

$\therefore P(0)$ holds

Inductive step:

We need to prove induction hypothesis implies $P(n++)$.

$$\text{If } n=0, \quad n++ = 0++$$

$$\text{If } n=0++, \quad n++ = (0++)++ \text{ which is of form } (m++)++ \text{ for } m=0.$$

$$\text{If } n=(m++)++, \quad n++ = (m++)++++) \text{ which is of form } (l++)++ \text{ for } l=m++.$$

$\therefore n++$ can be represented as the above forms.

$$\therefore P(n) \Rightarrow P(n++)$$

$$\therefore n=0 \text{ or } n=0++ \text{ or } n=(m++)++ \text{ from } m \in \mathbb{N}, \forall n \in \mathbb{N}.$$

$$\text{If } n=(m++)++, \quad m \text{ is unique.}$$

Proof: Let, $n=(m++)++$

Then, $(m++)++ = (g++)++$
 $m++ = g++$
 $m = g$

(d) Recursion theorem is generally of shape:

$$f(0) = a \quad \text{and} \quad f(n++) = F(n, f(n))$$

That is, we calculate $f(n++)$ using $f(n)$. - only 1 ^{function} step is required.

But here, we are ~~not~~ calculating $f(n++)++$ using $f(n)$, which is 2 steps backwards.

$\therefore$ eq (1) is not directly a case of recursion theorem.

(e) We can ^{use} a trick: recurse on pairs.

Value of h :

$$\begin{aligned} h(0) &= 0 \\ h(1) &= 0 \\ h(2) &= 1 \\ h(3) &= 1 \\ &\vdots \end{aligned}$$

We try to construct a g st:

$$\begin{aligned} g(0) &= (0, 0) = (h(0), h(1)) \\ g(1) &= (0, 1) = (h(1), h(2)) \\ g(2) &= (1, 1) = (h(2), h(3)) \\ g(3) &= (1, 2) = (h(3), h(4)) \\ &\vdots \end{aligned}$$

Reason: We can use $g(n)$ for generating $g(n++)$.

Formally: $g(0) = (0, 0)$

$$F: N \times N \rightarrow N \times N \quad F(a, b) = (a, a++)$$

$$g(n++) := F(g(n))$$

This is valid recursion theorem. — IMPORTANT

Informally, $g(n++) =$ (2nd component of $g(n)$, ~~the~~ 1st component of $g(n)$ plus 1).

Define $c_1: N \times N \rightarrow N$ to get first component of a pair as

$$c_1(a, b) = a$$

Then, we can say, $h = c_1 \circ g$

Informally, $h(n) =$ first component of $g(n) \quad \forall n \in N$.

We can check:

$$h(0) = c_1(g(0)) = c_1(0, 0) = 0$$

$$h(0++) = c_1(g(0++)) = c_1(0, 0++) = 0$$

$$\begin{aligned} h(n++++) &= c_1(g(n++++)) = c_1(F(g(n+++))) \\ &= c_1(F(F(g(n++)))) \\ &= \cancel{h(n++)} \dots \\ &= h(n)++ \end{aligned}$$

(d) Let both h and h' satisfy eqn.

We can use induction

$$P(n): h(n) = h'(n)$$

$$P(0): h(0) = 0 = h'(0)$$

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$P(n++) : w$

$$\begin{aligned}h(n++) ++ &= h(n) ++ \\&= h'(n) ++ \\&= h'(n++) ++\end{aligned}$$

[ind. hyp.thesis]

$\therefore P(n) \rightarrow P(n++)$

$\therefore h' = h$

$\therefore h$ is single.